

JOONWON LEE

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SUMMARY

Computational statistician and Statistics Ph.D. candidate specializing in scalable likelihood inference, probabilistic and predictive modeling, and statistical diagnostics for large, dependent datasets. Builds reproducible Python/PyTorch workflows for data integration, model estimation, simulation, and validation; translates business questions into measurable statistical problems, evaluation metrics, and defensible results.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Statistical Computing: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, Linux, Git, AWS EC2, HPC/SLURM; SQL/MySQL (working knowledge); CPU/GPU computing, data restructuring, and reproducible pipelines

Computational Statistics and Modeling: Scalable likelihood approximation, Gaussian processes, MLE/GLS, numerical optimization, Monte Carlo simulation, statistical diagnostics, hypothesis testing, regression (GLM/GBM), machine learning, model selection and validation, uncertainty quantification, and missing-data analysis

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Translated a business monitoring question into explicit hypotheses and measurable decision rules, developing and validating a Sequential Probability Ratio Test for binary risk indicators.
- Modeled the cumulative log-likelihood-ratio process as a random walk and applied Brownian-motion approximation and dynamic-programming probability propagation to evaluate boundary-crossing probabilities, stopping-time distributions, Type I/II error, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM regression benchmarking, model validation, and performance evaluation on AWS EC2.
- Presented model diagnostics and feature-attribution findings as pricing and risk-segmentation recommendations, translating technical results into clear actions for business stakeholders.

COMPUTATIONAL STATISTICS RESEARCH

Ph.D. Dissertation Research - Scalable Statistical Computing and Model Diagnostics

Sep 2024 - Present

- Scalable Inference:** Developed an advection-aware Vecchia likelihood approximation for nonseparable space-time Gaussian processes, using ordered local conditioning for likelihood-based parameter estimation and uncertainty quantification with up to 145,008 observations per day.
- Data Integration and Integrity:** Built workflows for quality filtering, missingness tracking, time-dependent coordinate offsets, nearest-center matching, spatial-threshold regular-grid construction, and dependency-aware ordering for noisy satellite data.
- Statistical Diagnostics and Computing:** Developed scale- and frequency-resolved diagnostics that localize covariance misspecification and distinguish persistent dependence from high-frequency noise and acquisition artifacts; implemented Python/PyTorch pipelines with L-BFGS optimization, CPU/GPU and HPC execution, and C++/pybind11 integration.

Economic Policy Research - EITC and Household Labor Supply

Hanyang University

- Estimated the effect of the Earned Income Tax Credit on household labor supply using fixed-effects regression and propensity-score matching with observed household covariates.

ADDITIONAL

Coursework: Econometrics, Machine Learning, Probability Theory, Linear Algebra, Statistical Computing, Stochastic Processes, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics

Languages: Korean (Native); English (Fluent); Chinese (Basic)